

Sentiment Classification for Serbian Short Reviews: Comparing TF-IDF Logistic Regression and BERTiĆ Across Domains

Ana Stanojčić

Abstract

This paper investigates binary sentiment classification for Serbian short reviews using the SentiComments.SR dataset (Batanović et al., 2020). The study focuses on strongly polar examples and compares a classical TF-IDF plus Logistic Regression baseline with BERTiĆ, a transformer model for Bosnian, Croatian, Montenegrin, and Serbian. The models are trained on movie review comments and evaluated both in-domain, on held-out movie data, and out-of-domain on a books verification corpus. The results show that BERTiĆ clearly outperforms the baseline on both datasets. On the movie test set, BERTiĆ reaches 0.9358 accuracy and 0.9298 macro F1, compared with 0.7920 accuracy and 0.7439 macro F1 for the baseline. On the books test set, BERTiĆ also generalizes well, reaching 0.9286 accuracy and 0.9069 macro F1. In contrast, the baseline tends to overpredict the positive class and performs poorly on negative recall. In contrast, the baseline tends to overpredict the positive class and performs poorly on negative recall. A qualitative analysis with LIME highlights how BERTiĆ handles difficult negative examples more reliably than the baseline. Overall, the results indicate that, within this binary strong-sentiment setup, contextual transformer representations provide a stronger approach to Serbian sentiment classification than sparse lexical features.

1 Introduction

The goal of sentiment classification is to determine whether a text expresses a positive or negative opinion. It is an important task in sentiment analysis, with common applications in review analysis, opinion monitoring, and related downstream systems that use user-generated feedback (Liu and Zhang, 2012; Pang et al., 2002). Early work framed the task as classifying reviews by overall polarity, and it remains a standard benchmark problem in NLP (Pang et al., 2002; Liu and Zhang, 2012).

While sentiment classification has been studied extensively for English, Serbian remains much less resourced, especially for short and informal review-like texts (Batanović et al., 2020). Existing Serbian sentiment resources are comparatively limited, and the SentiComments.SR framework was introduced specifically to support sentiment annotation and analysis for Serbian short texts (Batanović et al., 2016, 2020).

Prior work has shown that classical lexical approaches such as bag-of-words, TF-IDF, and linear classifiers provide strong baselines for sentiment classification, including in resource-limited settings (Pang et al., 2002; Batanović et al., 2016). For Serbian specifically, Batanović et al. showed that reliable sentiment classification can be achieved with classical machine learning methods, which makes such models a meaningful baseline for this project (Batanović et al., 2016). More recent research has shown that transformer-based models often outperform sparse lexical models by capturing contextual information and semantic interactions more effectively (Ljubešić and Lauc, 2021; Joshi et al., 2019).

Therefore, the aim of this project is to build on this line of work and compare a classical sparse lexical baseline with a contextual transformer model for Serbian sentiment classification and to examine whether the stronger model also generalizes better across domains. To do this, the project studies binary sentiment classification for Serbian short reviews using the SentiComments.SR dataset (Batanović et al., 2020). The focus is on strongly polarized comments, which keeps the task well defined while still preserving some linguistic difficulty, including short informal phrasing and sarcastic variants. The project compares two modeling approaches: a classical lexical baseline based on TF-IDF features and Logistic Regression, and BERTiĆ, a transformer language model for Bosnian, Croatian, Montenegrin, and Serbian (Ljubešić and

Lauc, 2021).

The study addresses two research questions:

1. How much does a transformer model outperform a classical sparse lexical baseline on Serbian sentiment classification?
2. How well do the models generalize from movie reviews to book reviews without additional training?

To complement the quantitative evaluation, the project also includes a qualitative interpretability analysis using LIME (Ribeiro et al., 2016).

2 Theoretical Background

A standard approach to text classification represents each document as a sparse lexical vector and then trains a classifier on top of that representation. In sentiment classification, this often means a bag-of-words or TF-IDF representation combined with a linear classifier such as Logistic Regression (Batanović et al., 2016; Pang et al., 2002). This family of methods has long been used as a strong baseline for polarity classification (Pang et al., 2002; Batanović et al., 2016).

TF-IDF assigns a weight to each term based on how important it is in one document relative to the corpus. A common formulation is

$$\text{tfidf}(t, d) = \text{tf}(t, d) \cdot \log \frac{N}{\text{df}(t)},$$

where $\text{tf}(t, d)$ is the term frequency of token t in document d , $\text{df}(t)$ is the number of documents containing t , and N is the total number of documents. The intuition is that words that occur often in one document but not in many other documents receive higher weight. In sentiment classification, TF-IDF can emphasize polarity-bearing words and short expressions while reducing the influence of very common function words (Batanović et al., 2016).

Given a TF-IDF vector $\mathbf{x}$, Logistic Regression predicts the probability of a class using the sigmoid function

$$P(y = 1 \mid \mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{x} + b) = \frac{1}{1 + e^{-(\mathbf{w}^\top \mathbf{x} + b)}}.$$

Here, $\mathbf{w}$ is a learned weight vector and b is a bias term. For binary sentiment classification, the model learns which lexical features are associated with positive or negative labels. Logistic Regression is simple, efficient, and interpretable, but it largely

ignores word order and broader context. Because of this, it depends heavily on surface overlap and may struggle when sentiment depends on contextual interactions rather than isolated words (Pang et al., 2002; Batanović et al., 2016).

Transformer-based language models instead produce contextual representations of tokens and sequences. In self-attention, each token representation is updated using weighted interactions with other tokens in the same input. A standard attention computation is:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^\top}{\sqrt{d_k}} \right) V,$$

where Q , K , and V represent query, key, and value matrices. This allows the model to represent a word in relation to the surrounding context rather than as one fixed feature. Transformer models therefore capture semantic dependencies and interactions that bag-of-words approaches cannot model directly (Vaswani et al., 2017; Ljubešić and Lauc, 2021).

In sentiment classification, a pretrained transformer is usually fine-tuned for sequence classification by placing a prediction layer on top of the encoder output for the input sequence. Compared with sparse lexical models, transformer-based classifiers can better capture contextual meaning, polarity shifts, and interactions between words carrying sentiment (Ljubešić and Lauc, 2021; Joshi et al., 2019). Therefore, BERT is particularly relevant for Serbian downstream tasks, because it was pretrained for Bosnian, Croatian, Montenegrin, and Serbian (Ljubešić and Lauc, 2021).

To support qualitative analysis, this project uses LIME, a local interpretability method. For text classification, LIME generates perturbed versions of the input by randomly removing words, obtains predictions for these variants, and fits a simple local surrogate model around the original instance. The resulting feature weights indicate which tokens contribute the most toward or against the predicted class in that local neighborhood (Ribeiro et al., 2016).

3 Data Description

The main dataset used in this project is SentiComments.SR, a Serbian sentiment dataset of short comments (Batanović et al., 2020). The resource includes a main corpus of movie-related comments

together with separate verification corpora, including a book-related corpus. The dataset website describes three corpora: a main movie corpus with 3490 comments, a movie verification corpus with 464 comments, and a book verification corpus with 173 comments (Batanović, 2024).

For this project, only comments with strong sentiment labels were retained. In the SentiComments.SR annotation scheme, the sign indicates polarity, where ‘+’ marks positive sentiment and ‘-’ marks negative sentiment, while the additional symbols show further information such as sentiment strength, ambiguity, subjectivity, or sarcasm. In particular, the label ‘1’ denotes a strong and clear sentiment, and the suffix ‘s’ marks sarcasm (Batanović et al., 2020).

Accordingly, the labels +1 and +1s were sorted to the positive class, while -1 and -1s were sorted to the negative class. This produced a binary sentiment classification task that preserves strongly polarized examples while allowing sarcastic variants to remain part of the data. The choice was made to keep the classification problem clear and interpretable for the project while avoiding weaker or more ambiguous sentiment categories.

After filtering, the movie corpus contained 2,257 examples, while the books corpus contained 112 examples. The movie data was split into training, development, and test sets using stratified sampling, resulting in 1,579 training examples, 226 development examples, and 452 test examples. The books corpus was used only as an external test set to evaluate cross-domain generalization.

The class distribution was moderately imbalanced. In the movie data, about 64.3% of the examples were positive, while in the books data about 73.2% were positive. The books comments were also shorter on average than the movie comments. These differences make the books corpus useful as a small out-of-domain evaluation set.

4 Implementation of Methods

The first model was a TF-IDF plus Logistic Regression baseline. Each comment was represented using TF-IDF weighted unigram and bigram features, and a Logistic Regression classifier was trained on top of this representation. This setup was chosen as it is a standard and well-established baseline for sentiment classification, which makes it a meaningful point of comparison. Additionally, lexical TF-IDF features are often competitive on short re-

view texts, especially when sentiment is expressed through strongly polarized words or short phrases (Pang et al., 2002; Batanović et al., 2016).

The second model was BERTiĆ, a transformer model pretrained on Bosnian, Croatian, Montenegrin, and Serbian text (Ljubešić and Lauc, 2021). BERTiĆ was chosen because it is language-appropriate for Serbian and because contextual transformer models are expected to capture semantic interactions that would be missed by sparse lexical baselines. The model was fine-tuned as a binary sequence classifier with two output labels, negative and positive.

The model was trained on the movie training set and tuned on the development set. The maximum sequence length was set to 128 because the comments were relatively short, so a larger input window was not necessary. The learning rate was set to 2×10^{-5} and training ran for three epochs, which is a common fine-tuning configuration for transformer classification tasks. These settings were intended to keep the experiment computationally manageable while still allowing the model to adapt effectively to the sentiment classification task.

Both models were first evaluated in-domain on the movie test set. They were then evaluated out-of-domain on the books verification corpus without retraining. This design was chosen to test not only raw performance on held-out data from the same domain, but also performance under domain shift. Since the books corpus differs from the movie corpus in topic, average length, and class balance, it provides a compact but meaningful external test of generalization.

Performance was measured using accuracy and macro F1. Accuracy gives an overall summary of correct predictions, while macro F1 is especially important in this study because the class distribution is imbalanced and because the ability to classify both positive and negative comments matters for interpretation. In addition, recall values for a specific class were considered to determine whether either model was biased toward the majority positive class (Opitz, 2024).

For qualitative analysis, two representative examples were selected for LIME. The first was an in-domain movie example where the baseline predicted the wrong label and BERTiĆ predicted the correct one: "Najgori film koji sam pogledala u životu" (The worst movie I have seen in my life). The second was an out-of-domain books example, "Film je ništa u odnosu na knjigu." (The movie is

nothing compared to the book), which showed the same pattern. These examples were chosen because they illustrate both the performance gap between the models and the performance of BERTić under domain shift.

5 Results

The main quantitative results are summarized below in-text, while the full metric table is provided in Appendix. On the movie test set, the TF-IDF plus Logistic Regression baseline achieved an accuracy of 0.7920. While, the accuracy value of 0.9358 has significantly improved by using BERTić.

On the books test set, the baseline achieved 0.8125 accuracy and 0.7257 macro F1, while BERTić again performed much better, with 0.9286 accuracy and 0.9069 macro F1. These results show that BERTić clearly outperformed the baseline on both in-domain and out-of-domain evaluation.

The class-level analysis revealed an important difference between the models. The baseline struggled particularly on negative examples. On the movie test set, negative recall for the baseline was 0.5031, while positive recall was 0.9519. A similar pattern appeared on the books test set, where negative recall was 0.4667 and positive recall was 0.9390. In contrast, BERTić was much more balanced. On the movie test set, it achieved 0.9006 recall for the negative class and 0.9553 for the positive class. On the books test set, the corresponding recalls were 0.8333 and 0.9634. Thus, the advantage of BERTić was not only overall, but especially noticeable for the negative class.

Generalization from movies to books was relatively stable for both models. The baseline showed only a small decrease in macro F1, from 0.7439 to 0.7257. BERTić also showed only a small drop, from 0.9298 to 0.9069. Although the books corpus differs from the movie corpus in length and class balance, the transformer model still produced very similar results.

6 Discussion of Results

The results show a clear advantage for contextual transformer models in Serbian sentiment classification. BERTić consistently outperformed the TF-IDF plus Logistic Regression baseline on both the in-domain movie test set and the out-of-domain books test set. The largest practical difference appeared in the treatment of negative examples. The baseline frequently overpredicted the positive

class, which greatly reduced negative recall, while BERTić remained much more balanced across both sentiment classes. This suggests that sparse lexical features are less reliable than contextual representations when sentiment is decided from short and variable review comments.

These findings are consistent with previous research showing that contextual transformer models generally outperform sparse lexical baselines when semantic context plays an important role (Joshi et al., 2019; Ljubešić and Lauc, 2021).

The qualitative LIME analysis supports this interpretation. In the movie example “Najgori film koji sam pogledala u životu!”, the baseline predicted positive sentiment, while BERTić correctly predicted negative sentiment. The LIME explanation showed that BERTić assigned strong importance to the explicitly negative token “Najgori” (the worst), whereas the baseline was influenced by other lexical patterns that weakened the negative interpretation.

LIME Explanation - baseline - In-domain example where baseline fails and BERTić succeeds
Prediction: positive | P(negative)=0.401, P(positive)=0.599
Text: Najgori film koji sam pogledala u životu!

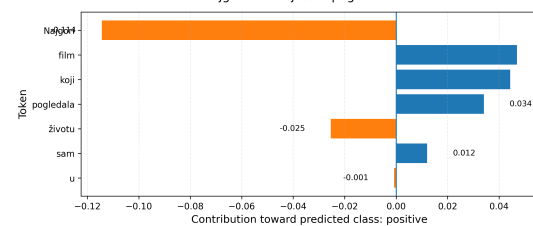


Figure 1: LIME explanation for the movie example “Najgori film koji sam pogledala u životu!” The TF-IDF + Logistic Regression baseline predicts the wrong positive label.

LIME Explanation - bertic - In-domain example where baseline fails and BERTić succeeds
Prediction: negative | P(negative)=0.818, P(positive)=0.182
Text: Najgori film koji sam pogledala u životu!

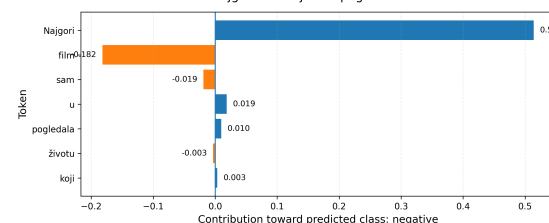


Figure 2: LIME explanation for the movie example “Najgori film koji sam pogledala u životu!” BERTić predicts the correct negative label and assigns strong importance to the explicitly negative cue “Najgori”.

A similar pattern appeared in the books example “Film je ništa u odnosu na knjigu.”, where the TF-IDF + Logistic Regression baseline predicted the

positive class with a probability of 0.5024, while BERTiC predicted the correct negative class with a probability of 0.8194. The example is particularly useful because it comes from the out-of-domain books corpus and still reveals the same difference between the models. The baseline appears to remain close to the decision boundary, whereas BERTiC captures the negative comparison more reliably. In particular, the negative meaning is expressed through the comparative construction “ništa u odnosu na knjigu,” (nothing in comparison to the book) which frames the film as clearly inferior to the book. This suggests that BERTiC is better able to use the contextual relation between words, rather than relying mainly on surface lexical cues. This supports the observation that the transformer model is better suited for generalization across the two domains considered in this study.

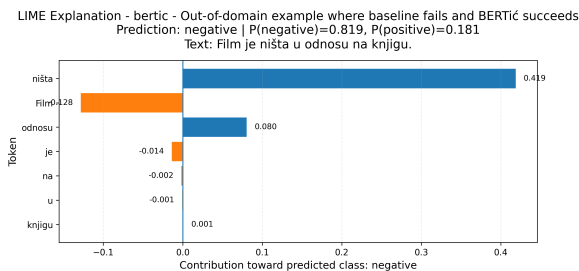


Figure 3: LIME explanation for the books example “Film je ništa u odnosu na knjigu.” BERTiC predicts the correct negative label and captures the negative comparison in this out-of-domain case.

Taken together, the quantitative and qualitative findings indicate that BERTiC provides more consistent performance for Serbian review sentiment classification than a classical TF-IDF based baseline. At the same time, the findings should be interpreted within the scope of the experimental setup and the data used..

7 Limitations

This study has several limitations. First, the results for cross-domain evaluation should be approached with caution as the books verification corpus is relatively small. While the out-of-domain results look promising, a larger and more diverse external test set would provide a stronger basis for conclusions about generalization.

Second, the project only uses comments with strong sentiment labels. This simplifies the task, but it excludes weaker, mixed, or more ambiguous sentiment categories. As a result, the findings

cannot be directly generalized to the full range of sentiment labels present in the original dataset.

Third, the qualitative interpretability analysis is limited to a small number of selected examples. The LIME explanations are useful for illustrating model behavior, but they should not be taken as a complete account of how the models behave across the full dataset.

Finally, the models were trained only on movie comments and evaluated on one additional domain (book comments). Other domains of Serbian short text may contain different lexical or stylistic properties, so further experiments would be needed to assess broader performance.

8 Conclusion

This paper examined binary sentiment classification for Serbian short reviews using SentiComments.SR. A TF-IDF plus Logistic Regression baseline was compared with BERTiC, a transformer model for Serbian, Bosnian, Croatian and Montenegrin. Across the experimental setup used here, BERTiC clearly outperformed the baseline on both movie and book reviews, with especially large gains in macro F1 and in recall for negative examples.

The cross-domain evaluation further suggested that BERTiC transfers well from movies to books, despite differences in average text length and class balance. The qualitative LIME analysis also indicated that BERTiC handles difficult negative cases more reliably than the lexical baseline, both in-domain and out-of-domain.

Within the scope of this binary sentiment setup, the project shows that contextual transformer models provide a substantially stronger approach to Serbian sentiment classification than classical sparse lexical baselines. Future work could extend the analysis to the full SentiComments.SR label set, broader domain coverage, and more comprehensive qualitative evaluation.

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Appendix: Full Results Table

Main Results on Movie and Book Test Sets

Model	Dataset	Accuracy	Macro F1	Neg. Precision	Neg. Recall	Pos. Precision	Pos. Recall
DF + Logistic Regress	Movies	0.7920	0.7439	0.8526	0.5031	0.7759	0.9519
DF + Logistic Regress	Books	0.8125	0.7257	0.7368	0.4667	0.8280	0.9390
BERTiĆ	Movies	0.9358	0.9298	0.9177	0.9006	0.9456	0.9553
BERTiĆ	Books	0.9286	0.9069	0.8929	0.8333	0.9405	0.9634

Figure 1: Performance of TF-IDF + Logistic Regression and BERTiĆ on the movie and books test sets.